

ECE 456 Problem Set 1:

The Basics of Electron Flow in Nanodevices

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1 Current with a Lorentzian Density of States

1.1 Alternative Form

Given the following equations for the number of channel electrons and current in a nanodevice

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E) + \gamma_2 f_2(E)}{\gamma_1 + \gamma_2} D(E - U) dE \quad (1)$$

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{P} \gamma_1 + \gamma_2 \int_{-\infty}^{\infty} [f_1(E) - f_2(E)] D(E - U) dE \quad (2)$$

These equations can be manipulated to be easier to code into MATLAB. Firstly, change the variable of integration to $E' = E - U$. This can be manipulated to receive

$$E' = E - U$$

$$E = E' + U$$

$$dE' = dE$$

Originally $E : -\infty \rightarrow \infty$, however, since subtracting a constant doesn't change infinity

$$E' : -\infty \rightarrow \infty$$

Hence, substituting for $E = E' + U$ and replacing $E' \rightarrow E$ produces

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E + U) + \gamma_2 f_2(E + U)}{\gamma_1 + \gamma_2} D(E) dE \quad (3)$$

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} \int_{-\infty}^{\infty} [f_1(E + U) - f_2(E + U)] D(E) dE \quad (4)$$

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